Subharmonicity on All Affine Complex Lines Forces Upper Semicontinuity

Abstract

Let $\Omega \subset \mathbb{C}^n$ be a domain and let $\phi\colon \Omega \to [-\infty, \infty)$ be not identically $-\infty$. We prove that if ϕ is subharmonic or identically $-\infty$ on every connected component of every affine complex-line section of Ω, then ϕ is automatically upper semicontinuous. Thus condition (2) in Definition 1.1.2 of [Xia] is redundant.

1 Statement and auxiliary lemmas

Theorem 1.1. *Let $\Omega \subset \mathbb{C}^n$ be a domain and let $\phi\colon \Omega \to [-\infty, \infty)$ be not identically $-\infty$. Suppose that for every affine complex line $L \subset \mathbb{C}^n$ and every connected component U of $L \cap \Omega$, the restriction $\phi|_U$ is subharmonic or identically $-\infty$. Then ϕ is upper semicontinuous, hence plurisubharmonic.*

Throughout, a function u on an open set $D \subset \mathbb{C}^n$ is said to be *subharmonic on every affine complex-line section* of D if for every affine complex line L and every connected component U of $L \cap D$, the restriction $u|_U$ is subharmonic (in particular finite somewhere and upper semicontinuous on U). For a finite Borel measure μ on $\mathbb{C}^n$ and $0 < s < 2n$ we write

$$I_s(\mu) = \iint |x-y|^{-s}\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y) \in [0, \infty]$$

for its Riesz s-energy. All constants denoted $c_n, C_{n,R}, \dots$ depend only on the indicated parameters.

The first lemma replaces the measurability lemma of Arsove [Ars66, Lemma 1]. Arsove's lemma yields Lebesgue measurability of doubly subharmonic functions; we need *Borel* measurability in Lemma 1.3 below, so we include the short argument.

Lemma 1.2 (Measurability)**.** *Let $n \geq 2$, let $G \subset \mathbb{C}^{n-1} \times \mathbb{C}$ be open, and let $u\colon G \to [0, \infty)$ satisfy:*

- (i) *for every $w \in \mathbb{C}$, the function $z' \mapsto u(z', w)$ is upper semicontinuous on $\{z' : (z', w) \in G\}$;*
- (ii) *for every $z' \in \mathbb{C}^{n-1}$, the function $w \mapsto u(z', w)$ is subharmonic on every connected component of $\{w : (z', w) \in G\}$.*

Then u is Borel measurable.

Proof. Borel measurability is a local property, so it suffices to show that u is Borel on $B' \times B$ whenever $B' \subset \mathbb{C}^{n-1}$ is an open ball and $B = B(w_0, \rho) \subset \mathbb{C}$ is a disc with $B' \times B(w_0, 2\rho) \subset G$. Fix such B', B. For $N \in \mathbb{N}$ and $0 < r \leq \rho$ put $u^N = \min\{u, N\}$ and

$$F_r^N(z', w) = \frac{1}{\pi r^2} \int_{B(w,r)} u^N(z', \eta)\, \mathrm{d}A(\eta), \qquad (z', w) \in B' \times B.$$

This makes sense: the disc $B(w_0, 2\rho)$ is connected and contained in $\{w : (z', w) \in G\}$, so $u(z', \cdot)$ is subharmonic, in particular upper semicontinuous, on it by (ii); moreover $B(w, r) \subset B(w_0, 2\rho)$ and $0 \leq u^N \leq N$.

For fixed w, the function $F_r^N(\cdot, w)$ is upper semicontinuous on B': if $z'_k \to z'$ in B', then the reverse Fatou lemma (the integrands are bounded above by N on a set of finite measure) and (i) give

$$\limsup_{k\to\infty} F_r^N(z'_k, w) \le \frac{1}{\pi r^2} \int_{B(w,r)} \limsup_{k\to\infty} u^N(z'_k, \eta)\, \mathrm{d}A(\eta) \le F_r^N(z', w).$$

For fixed z', the function $F_r^N(z', \cdot)$ is continuous on B, since $|F_r^N(z', w) - F_r^N(z', w')| \le N\, |B(w, r) \triangle B(w', r)|/(\pi r^2)$. A function which is Borel in the first variable and continuous in the second is jointly Borel: if for each k we partition B into countably many Borel sets $Q_{k,i}$ of diameter less than $1/k$ and pick $w_{k,i} \in Q_{k,i}$, then $F_r^N(z', w) = \lim_k \sum_i F_r^N(z', w_{k,i}) \mathbf{1}_{Q_{k,i}}(w)$, and each function on the right is Borel. Hence F_r^N is Borel on $B' \times B$, and so is

$$G^N(z', w) := \limsup_{k\to\infty} F^N_{\rho/k}(z', w).$$

Now fix $(z', w) \in B' \times B$. By (ii), $u(z', \cdot)$ is subharmonic on the disc $B(w_0, 2\rho)$, hence bounded above on the compact disc $\overline{B}(w, \rho)$, say by N_0. For $N \ge N_0$ we have $u^N(z', \cdot) = u(z', \cdot)$ on $B(w, \rho)$, so $F^N_{\rho/k}(z', w)$ is the mean of the subharmonic function $u(z', \cdot)$ over $B(w, \rho/k)$, which converges to $u(z', w)$ as $k \to \infty$. Thus $G^N(z', w) = u(z', w)$ for all $N \ge N_0$, and $u = \lim_{N\to\infty} G^N$ is Borel. □

Lemma 1.3 (High-value star)**.** *Let $n \ge 2$, let $D \subset \mathbb{C}^n$ be open with $\overline{B}(x, R) \subset D$, and let $u\colon D \to \mathbb{R}$ be Borel and subharmonic on every affine complex-line section of D. Put $A = u(x)$. Then there is a Borel probability measure μ on $\mathbb{C}^n$ carried by*

$$E(x, A, R) = \{y \in \mathbb{C}^n : R/2 \le |y - x| \le R,\ u(y) \ge A\}$$

such that

$$I_{2n-2}(\mu) \le C_{n,R}. \tag{1.1}$$

The constant $C_{n,R}$ is independent of u, x and A.

Proof. Let $P = \mathbb{CP}^{n-1}$ with its $U(n)$-invariant probability measure σ and Fubini–Study distance $d(\xi, \xi') = \arccos |\langle e, e'\rangle| \in [0, \pi/2]$, where e, e' are unit vectors spanning ξ, ξ', and let $q\colon S^{2n-1} \to P$ be the Hopf map $q(e) = [e]$.

Fix $r \in [R/2, R]$ and $\xi \in P$, and pick a unit vector e_0 spanning ξ. The closed disc $\{x + te_0 : |t| \le R\}$ lies in $\overline{B}(x, R) \subset D$, hence in a single connected component of $(x + \mathbb{C}e_0) \cap D$, on which u is subharmonic. In particular $\theta \mapsto u(x + re^{\mathrm{i}\theta} e_0)$ is upper semicontinuous, and the submean inequality gives

$$A = u(x) \le \frac{1}{2\pi} \int_0^{2\pi} u(x + re^{\mathrm{i}\theta} e_0)\, \mathrm{d}\theta.$$

Consequently the set

$$\Gamma_{(r,\xi)} = \{e \in q^{-1}(\xi) : u(x + re) \ge A\}$$

is closed in the circle $q^{-1}(\xi)$, hence compact, and it is nonempty: otherwise the upper semicontinuous function u would have maximum strictly less than A on the circle $\{x + re : e \in q^{-1}(\xi)\}$, contradicting the submean inequality. Since u is Borel, the set

$$\Gamma = \{(r, \xi, e) \in [R/2, R] \times P \times S^{2n-1} : q(e) = \xi,\ u(x + re) \ge A\}$$

is Borel, and its sections $\Gamma_{(r,\xi)}$ are nonempty and compact. By the Arsenin–Kunugui uniformization theorem [Kec95, Theorem 18.18] there is a Borel map $e\colon [R/2, R] \times P \to S^{2n-1}$ with $(r, \xi, e(r, \xi)) \in \Gamma$ for all (r, ξ).

Let m be the product of normalized Lebesgue measure on $[R/2, R]$ with σ, let $\Psi(r, \xi) = x + r\, e(r, \xi)$, and put $\mu = \Psi_\# m$. Then μ is a Borel probability measure carried by $E(x, A, R)$.

It remains to bound the energy. For unit vectors e, e' spanning ξ, ξ' and $r, r' \in [R/2, R]$,

$$|re - r'e'|^2 = (r - r')^2 + rr'|e - e'|^2 \geq (r - r')^2 + \frac{R^2}{4}(2 - 2|\langle e, e'\rangle|),$$

and $2 - 2|\langle e, e'\rangle| = 4\sin^2(d(\xi, \xi')/2) \geq \frac{4}{\pi^2} d(\xi, \xi')^2$ because $\sin s \geq 2s/\pi$ on $[0, \pi/2]$. Hence

$$|\Psi(r, \xi) - \Psi(r', \xi')|^2 \geq (r - r')^2 + \frac{R^2}{\pi^2} d(\xi, \xi')^2 \geq c_R\, \varrho\big((r, \xi), (r', \xi')\big)^2,$$

where ϱ, given by $\varrho^2 = (r - r')^2 + d(\xi, \xi')^2$, is the product metric on $[R/2, R] \times P$ and $c_R = \min\{1, R^2/\pi^2\}$. Therefore

$$I_{2n-2}(\mu) \leq c_R^{-(n-1)} \iint \varrho(s, t)^{-(2n-2)}\, \mathrm{d}m(s)\, \mathrm{d}m(t).$$

The space $[R/2, R] \times P$ is a compact Riemannian manifold with boundary of real dimension $2n - 1$, ϱ is its Riemannian distance, and m is a constant multiple of its Riemannian volume. Cover the space by finitely many charts and let $\varepsilon > 0$ be a Lebesgue number of the cover: pairs (s, t) with $\varrho(s, t) \geq \varepsilon$ contribute at most $\varepsilon^{-(2n-2)}$ to the last integral, while pairs with $\varrho(s, t) < \varepsilon$ lie in a common chart, in which ϱ is comparable to the Euclidean distance. Hence the last integral is bounded by a constant depending only on n and R times $1 + \int_{|s|<1} |s|^{-(2n-2)}\, \mathrm{d}s$, the integral being over the unit ball of $\mathbb{R}^{2n-1}$; it is finite because $2n - 2 < 2n - 1$. This proves (1.1). ☐

Let $\mathcal{H} = \mathrm{Gr}_{\mathbb{C}}(n - 1, n)$ be the Grassmannian of complex hyperplanes $H \subset \mathbb{C}^n$ through the origin, with its $U(n)$-invariant probability measure ν. For $H \in \mathcal{H}$ let P_H be the orthogonal projection onto H and m_H Lebesgue measure on H (real dimension $2n - 2$). For a finite Borel measure μ on $\mathbb{C}^n$ put $\mu_H = (P_H)_\# \mu$. We use the Fourier transform $\widehat{\mu}(\zeta) = \int e^{-2\pi \mathrm{i}\, \mathrm{Re}\langle y, \zeta\rangle}\, \mathrm{d}\mu(y)$, where $\langle \cdot, \cdot \rangle$ is the Hermitian inner product of $\mathbb{C}^n$; the same convention is used on each H. For a Borel set $S \subset \mathbb{C}^n$ the function $H \mapsto m_H(S \cap H)$ is Borel on $\mathcal{H}$ (this is clear for open S and follows in general by a monotone class argument), which justifies the integrals over $\mathcal{H}$ below.

Lemma 1.4 (Projections onto complex hyperplanes)**.** *Let μ be a compactly supported finite Borel measure on $\mathbb{C}^n$, $n \geq 2$, with $I_{2n-2}(\mu) < \infty$. Then for ν-almost every $H \in \mathcal{H}$ the measure μ_H is absolutely continuous with respect to m_H, with density $g_H \in L^2(H, m_H)$, and*

$$\int_{\mathcal{H}} \|g_H\|^2_{L^2(H)}\, \mathrm{d}\nu(H) = c_n\, I_{2n-2}(\mu). \tag{1.2}$$

Proof. This is the analogue, for complex hyperplanes, of [Mat95, Theorem 9.7], and the proof is the same. First, for every Borel function $g \geq 0$ on $\mathbb{C}^n$,

$$\int_{\mathcal{H}} \int_H g\, \mathrm{d}m_H\, \mathrm{d}\nu(H) = a_n \int_{\mathbb{C}^n} g(\zeta)|\zeta|^{-2}\, \mathrm{d}\zeta. \tag{1.3}$$

Indeed, the left-hand side defines a Radon measure τ on $\mathbb{C}^n$ with $\tau(\{0\}) = 0$; it is $U(n)$-invariant, because ν is and $U_\# m_H = m_{UH}$ for $U \in U(n)$, and it satisfies $\tau(tE) = t^{2n-2}\tau(E)$ for $t > 0$. Since $U(n)$ acts transitively on the unit sphere, in polar coordinates τ is the product of a measure on $(0, \infty)$ with the normalized surface measure on S^{2n-1}, and homogeneity forces the radial factor to be a constant multiple of $t^{2n-3}\, \mathrm{d}t$. As Lebesgue measure is $t^{2n-1}\, \mathrm{d}t$ times surface measure, (1.3) follows.

Second, for $\zeta \in H$ we have $\langle P_H y, \zeta\rangle = \langle y, \zeta\rangle$ because $y - P_H y \in H^\perp$, so $\widehat{\mu_H} = \widehat{\mu}|_H$. Apply (1.3) to $g = |\widehat{\mu}|^2$ and use the Fourier representation of the energy [Mat95, Lemma 12.12], $I_{2n-2}(\mu) = b_n \int |\widehat{\mu}(\zeta)|^2 |\zeta|^{-2}\, \mathrm{d}\zeta$, to get

$$\int_{\mathcal{H}} \int_H |\widehat{\mu}|^2\, \mathrm{d}m_H\, \mathrm{d}\nu(H) = a_n \int_{\mathbb{C}^n} |\widehat{\mu}(\zeta)|^2 |\zeta|^{-2}\, \mathrm{d}\zeta = \frac{a_n}{b_n} I_{2n-2}(\mu) < \infty.$$

Hence for ν-almost every H the finite measure μ_H has Fourier transform in $L^2(H)$; by Plancherel's theorem it is then absolutely continuous with a density $g_H \in L^2(H)$ satisfying $\|g_H\|^2_{L^2(H)} = \int_H |\widehat{\mu_H}|^2 \,\mathrm{d}m_H$. Integrating over $\mathcal{H}$ gives (1.2). $\square$

Lemma 1.5 (Incidence)**.** *Let $n \geq 2$, let $K \subset \mathbb{C}^n$ be compact, and let $E_j \subset K$ ($j \geq 1$) be Borel sets. Suppose there are Borel probability measures μ_j with*

$$\mu_j(E_j) = 1, \qquad I_{2n-2}(\mu_j) \leq C \quad \textit{for all } j.$$

Then some affine complex line meets E_j for infinitely many j.

Proof. Choose M with $K \subset \overline{B}(0, M)$ and put

$$\mathcal{L} = \{(H, c) : H \in \mathcal{H},\ c \in H,\ |c| \leq M\},$$

a compact subset of $\mathcal{H} \times \mathbb{C}^n$; the pair (H, c) encodes the affine complex line $c + H^\perp$. Let λ be the finite measure on $\mathcal{L}$ given by

$$\int_{\mathcal{L}} h \,\mathrm{d}\lambda = \int_{\mathcal{H}} \int_{H \cap \overline{B}(0,M)} h(H, c) \,\mathrm{d}m_H(c) \,\mathrm{d}\nu(H), \qquad h \geq 0 \text{ Borel on } \mathcal{L}.$$

Fix j. By inner regularity there is a compact set $F_j \subset E_j$ with $\mu_j(F_j) \geq 1/2$. Let

$$\mathcal{I}_j = \{(H, c) \in \mathcal{L} : c \in P_H(F_j)\},$$

the image of the compact set $\mathcal{H} \times F_j$ under the continuous map $(H, y) \mapsto (H, P_H y)$; thus $\mathcal{I}_j$ is compact, its section over H is $P_H(F_j)$, and $(H, c) \in \mathcal{I}_j$ means precisely that the line $c + H^\perp$ meets F_j. Let $g_{j,H}$ be the densities provided by Lemma 1.4 for μ_j. Since $F_j \subset P_H^{-1}(P_H F_j)$, for ν-almost every H the Cauchy–Schwarz inequality gives

$$\frac{1}{2} \leq \mu_j(F_j) \leq (\mu_j)_H(P_H F_j) = \int_{P_H F_j} g_{j,H} \,\mathrm{d}m_H \leq \|g_{j,H}\|_{L^2(H)}\, m_H(P_H F_j)^{1/2},$$

that is, $m_H(P_H F_j) \geq \big(4\|g_{j,H}\|^2_{L^2(H)}\big)^{-1}$. Integrating over $\mathcal{H}$ and using Jensen's inequality for the convex function $t \mapsto 1/t$ together with (1.2),

$$\begin{aligned} \lambda(\mathcal{I}_j) = \int_{\mathcal{H}} m_H(P_H F_j) \,\mathrm{d}\nu(H) &\geq \int_{\mathcal{H}} \frac{\mathrm{d}\nu(H)}{4\|g_{j,H}\|^2_{L^2(H)}} \geq \frac{1}{4 \int_{\mathcal{H}} \|g_{j,H}\|^2_{L^2(H)} \,\mathrm{d}\nu(H)} \\ &= \frac{1}{4c_n I_{2n-2}(\mu_j)} \geq \frac{1}{4c_n C} =: \delta > 0. \end{aligned}$$

Since $\lambda(\mathcal{L}) < \infty$ and $\lambda(\bigcup_{j \geq m} \mathcal{I}_j) \geq \lambda(\mathcal{I}_m) \geq \delta$ for every m,

$$\lambda\Big(\bigcap_{m \geq 1} \bigcup_{j \geq m} \mathcal{I}_j\Big) = \lim_{m \to \infty} \lambda\Big(\bigcup_{j \geq m} \mathcal{I}_j\Big) \geq \delta > 0.$$

Any (H, c) in this set lies in $\mathcal{I}_j$ for infinitely many j, and then the line $c + H^\perp$ meets $F_j \subset E_j$ for infinitely many j. $\square$

2 Proof of the theorem

We first treat finite nonnegative functions.

Proposition 2.1. *Let $D \subset \mathbb{C}^n$ be a domain and let $u \colon D \to [0, \infty)$ be subharmonic on every affine complex-line section of D. Then u is plurisubharmonic.*

Note that the conclusion concerns the given function u itself, not merely its almost-everywhere class.

Proof. We argue by induction on n. For $n = 1$ the hypothesis says that u is subharmonic on D, which is the claim. Let $n \geq 2$ and assume the proposition in dimension $n-1$. Write $z = (z', w) \in \mathbb{C}^{n-1} \times \mathbb{C}$.

Step 1: u is Borel. Fix $w \in \mathbb{C}$. The set $D_w = \{z' : (z', w) \in D\}$ is open in $\mathbb{C}^{n-1}$, and on each of its connected components $u(\cdot, w)$ is subharmonic on every affine complex-line section, because an affine complex line of $\mathbb{C}^{n-1} \times \{w\}$ is an affine complex line of $\mathbb{C}^n$ and the connected components of the corresponding sections agree. By the induction hypothesis $u(\cdot, w)$ is plurisubharmonic, in particular upper semicontinuous, on D_w. For fixed z', the function $u(z', \cdot)$ is subharmonic on every connected component of $\{w : (z', w) \in D\}$ by hypothesis. Lemma 1.2 shows that u is Borel.

Step 2: u is locally bounded above. Otherwise there are $x_j \to p \in D$ with $A_j := u(x_j) \to \infty$. Choose $R > 0$ with $\overline{B}(p, 3R) \subset D$ and, after discarding finitely many terms, $|x_j - p| < R/2$. Then $\overline{B}(x_j, R) \subset D$, and Lemma 1.3 supplies Borel probability measures μ_j with $I_{2n-2}(\mu_j) \leq C_{n,R}$ carried by the Borel sets

$$E_j = \{y : R/2 \leq |y - x_j| \leq R,\ u(y) \geq A_j\} \subset K := \overline{B}(p, 2R).$$

By Lemma 1.5 there is an affine complex line L meeting E_j for infinitely many j; choose $y_j \in L \cap E_j$ for those j. The compact set $L \cap K$ is contained in the disc $L \cap B(p, 3R)$, hence in a single connected component U of $L \cap D$, on which u is subharmonic and therefore bounded above on compact subsets. But $y_j \in L \cap K$ and $u(y_j) \geq A_j \to \infty$, a contradiction.

Step 3: $\mathrm{dd^c} u \geq 0$. By Steps 1 and 2, u is a nonnegative locally bounded Borel function, so $u \in L^1_{\mathrm{loc}}(D)$. Fix $a \in \mathbb{C}^n \setminus \{0\}$ and a test function $\chi \geq 0$ on D. Writing points as $z = v + ta$ with $v \in (\mathbb{C}a)^\perp$ and $t \in \mathbb{C}$, we have $\sum_{k,\ell} a_k \overline{a_\ell}\, \partial^2_{z_k \overline{z}_\ell} \chi(v + ta) = \partial^2_{t\overline{t}}[\chi(v + ta)]$, so by Fubini's theorem

$$\Big\langle \sum_{k,\ell=1}^{n} a_k \overline{a_\ell} \frac{\partial^2 u}{\partial z_k \partial \overline{z}_\ell}, \chi \Big\rangle = |a|^2 \int_{(\mathbb{C}a)^\perp} \Big(\int_{\mathbb{C}} u(v + ta)\, \partial^2_{t\overline{t}}[\chi(v + ta)]\, \mathrm{d}A(t) \Big) \mathrm{d}v \geq 0,$$

because for each v the function $t \mapsto u(v + ta)$ is subharmonic on each connected component of the open set $O_v = \{t : v + ta \in D\}$, and the nonnegative test function $t \mapsto \chi(v + ta)$ on O_v has compact support, hence meets only finitely many components, on each of which the inner integral is nonnegative. Thus $T_a := \sum_{k,\ell} a_k \overline{a_\ell}\, \partial^2_{z_k \overline{z}_\ell} u$ is a positive distribution, hence a positive measure, for every $a \in \mathbb{C}^n$. Taking $a = e_k$, $a = e_k + e_\ell$ and $a = e_k + \mathrm{i} e_\ell$ shows that each coefficient $\partial^2_{z_k \overline{z}_\ell} u$ is a complex measure, and the positivity of all T_a says exactly that the matrix of these measures is positive semidefinite, i.e. $\mathrm{dd^c} u \geq 0$ in the sense of currents. By the distributional characterization of plurisubharmonic functions [Xia, Theorem 1.1.2] there is $\widetilde{u} \in \mathrm{PSH}(D)$ with $u = \widetilde{u}$ almost everywhere.

Step 4: $u = \widetilde{u}$ everywhere. Fix $z \in D$ and $\rho > 0$ with $B(z, \rho) \subset D$, and let $N = \{u \neq \widetilde{u}\}$, a Borel set of Lebesgue measure zero. In the polar coordinates $y = z + tv$, $t \in \mathbb{C}$, $[v] \in \mathbb{CP}^{n-1}$ (v a unit vector), Lebesgue measure is $C_n |t|^{2n-2}\, \mathrm{d}A(t)\, \mathrm{d}\sigma([v])$, so

$$0 = |N \cap B(z, \rho)| = C_n \int_{\mathbb{CP}^{n-1}} \Big(\int_{|t| < \rho} \mathbf{1}_N(z + tv) |t|^{2n-2}\, \mathrm{d}A(t) \Big) \mathrm{d}\sigma([v]).$$

Hence for σ-almost every direction $[v]$ the set $N \cap (z + \mathbb{C}v)$ has planar measure zero in the disc $\{z + tv : |t| < \rho\}$. On such a disc $t \mapsto u(z + tv)$ is subharmonic by hypothesis, and $t \mapsto \widetilde{u}(z + tv)$ is subharmonic or identically $-\infty$ since $\widetilde{u}$ is plurisubharmonic; the latter alternative is excluded because $\widetilde{u} = u$ almost everywhere on the disc and u is finite. Two subharmonic functions which agree almost everywhere agree everywhere (each is the limit of its own disc means), so $u(z) = \widetilde{u}(z)$. Thus $u = \widetilde{u}$ is plurisubharmonic. $\square$

Proof of Theorem 1.1. For $m \geq 1$ set

$$\phi_m = \max\{\phi, -m\}, \qquad u_m = \phi_m + m.$$

The function u_m is finite and nonnegative. On each connected component U of an affine complex-line section of Ω, either $\phi|_U$ is subharmonic, and then so is $\phi_m|_U = \max\{\phi|_U, -m\}$, or $\phi|_U \equiv -\infty$, and then $\phi_m|_U \equiv -m$ is constant. Thus u_m satisfies the hypotheses of Proposition 2.1, and every ϕ_m is plurisubharmonic, in particular upper semicontinuous. Since $\phi \leq \phi_m$ and $\phi_m \downarrow \phi$ pointwise, for every $a \in \mathbb{R}$,

$$\{\phi < a\} = \bigcup_{m \geq 1} \{\phi_m < a\}.$$

The right-hand side is open, so ϕ is upper semicontinuous. Together with the assumption on line sections and $\phi \not\equiv -\infty$, this is exactly the definition of a plurisubharmonic function [Xia, Definition 1.1.2]. $\square$

Remark 2.2. Wiegerinck's example [Wie88] is separately subharmonic with respect to a fixed product splitting $\mathbb{C} \times \mathbb{C}$, but is not locally bounded above, so some hypothesis on other directions is needed. The all-directions hypothesis rules out its spike mechanism: Lemmas 1.3 and 1.5 force infinitely many spikes to be seen by one fixed affine complex line, on which they are incompatible with subharmonicity. Note also that the incidence geometry of non-coordinate lines enters only through Step 2 of Proposition 2.1: once u is known to be Borel and locally bounded above, Steps 3 and 4 are the standard passage from subharmonicity on lines to plurisubharmonicity, cf. [Ars66].

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